Deliverable #3 Template

SE 3A04: Software Design II – Large System Design

Tutorial Number: T01

Group Number: G06

Group Members:

- Aakash Satishkumar
- Ali Shareeff
- Harmanpreet Singh Sagar
- Jason Kim
- Praneet Singh

IMPORTANT NOTES

- You do NOT need to provide a text explanation of each diagram; the diagram should speak for itself
- Please document any non-standard notations that you may have used
    - *Rule of Thumb*: if you feel there is any doubt surrounding the meaning of your notations, document them
- Some diagrams may be difficult to fit into one page
    - It is OK if the text is small but please ensure that it is readable when printed
    - If you need to break a diagram onto multiple pages, please adopt a system of doing so and thoroughly explain how it can be reconnected from one page to the next; if you are unsure about this, please ask me
- Please submit the latest version of Deliverable 1 and Deliverable 2 with Deliverable 3
- They do not have to be a freshly printed versions; the latest marked versions are OK
- If you do NOT have a Division of Labour sheet, your deliverable will NOT be marked

1 Introduction

This section should provide a brief overview of the entire document.

1.1 Purpose

The purpose of this document is to provide a detailed class diagram, sequence diagrams, and state charts that illustrate the SCEMAS system architecture. By giving a more accurate representation of system behaviour, component interactions, and overall structure, it expands upon the framework created in earlier deliverables. This document is meant for SCEMAS internal stakeholders, such as developers, project managers, domain specialists, and investors. To properly interpret the diagrams and system design, it is assumed that one is familiar with Deliverables 1 and 2, as well as have a fundamental understanding of software design ideas.

1.2 System Description

SCEMAS is an environmental monitoring system that gathers, processes, and evaluates data from distributed sensors, including measurements of temperature, noise levels, and air quality. The system converts unprocessed sensor data into valuable information that helps different stakeholders make decisions. It continuously scans incoming data for abnormalities and threshold violations, sending out notifications when unusual circumstances are found. To facilitate trend analysis and long-term assessment of environmental patterns, SCEMAS keeps historical records in addition to real-time monitoring. Additionally, the system maintains thorough records of system events and alarms for accountability and traceability, and it guarantees that validated data is available via a public API.

1.3 Overview

This document presents a collection of diagrams that illustrate different aspects of the SCEMAS system design. State charts for the controller classes, which demonstrate the internal behaviour and transitions of significant system components, are included in Section 2. Sequence diagrams that illustrate component interactions and correlate to the system's business events are presented in Section 3. A comprehensive UML class diagram that depicts the general system structure and class interactions is shown in Section 4. The GitHub repository containing the source code for all the UML diagrams in this document is as follows: https://github.com/harmanpreet-sagar/SFWRENG-3A04-Group6-T01

2 State Charts for Controller Classes

This section should provide a state chart for each controller class for your application.

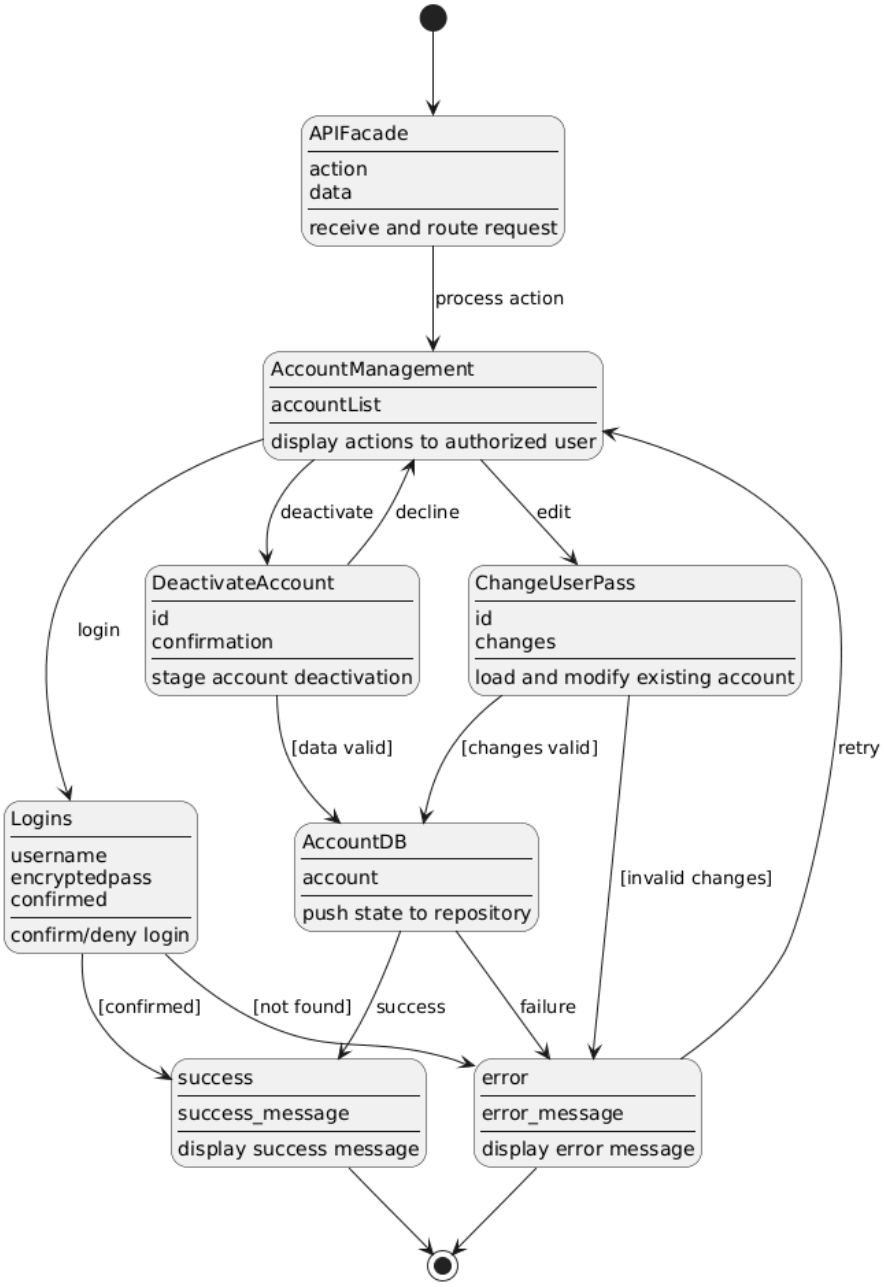

Figure 1: Account Management State Chart

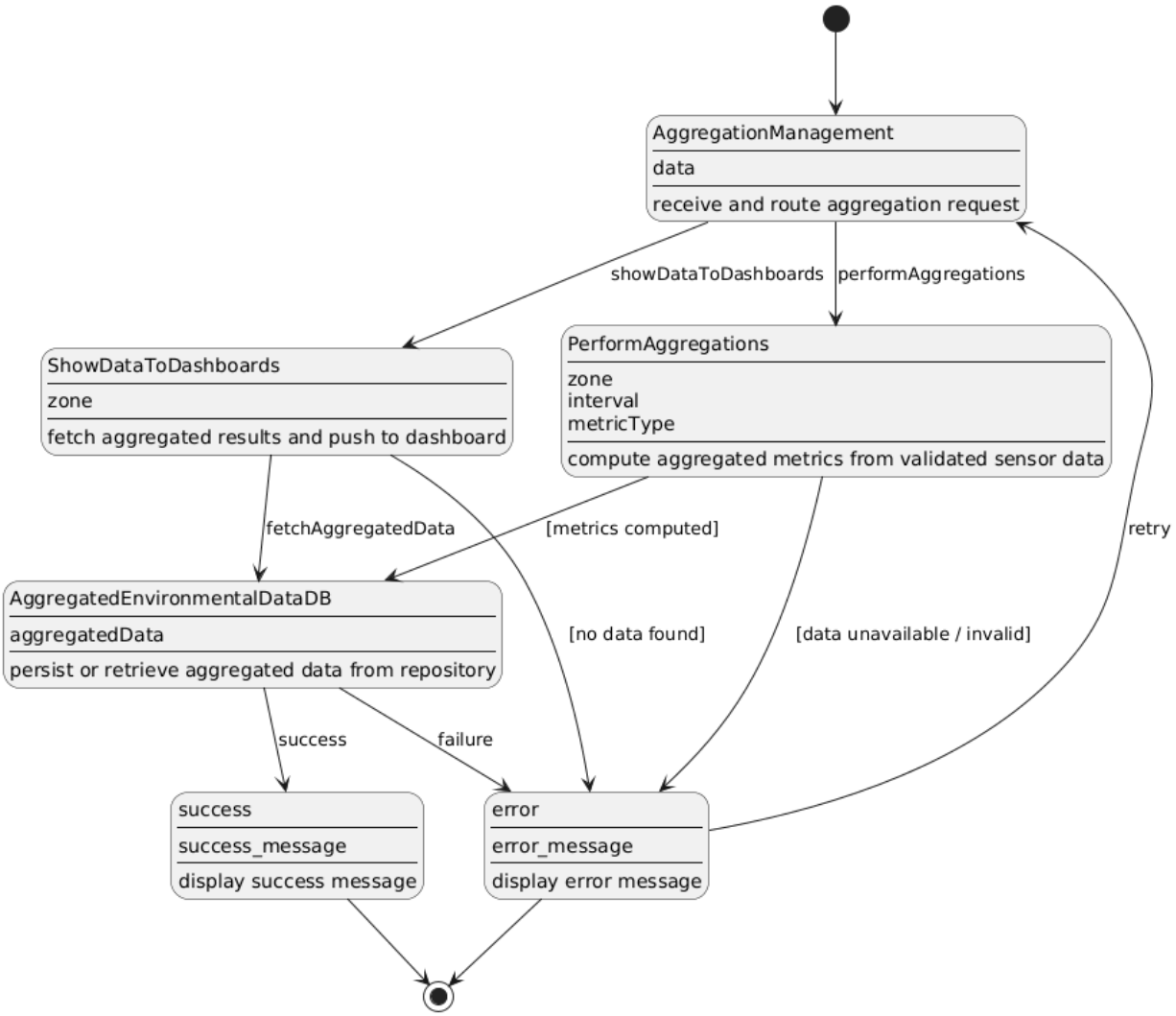

Figure 2: Aggregation Management State Chart

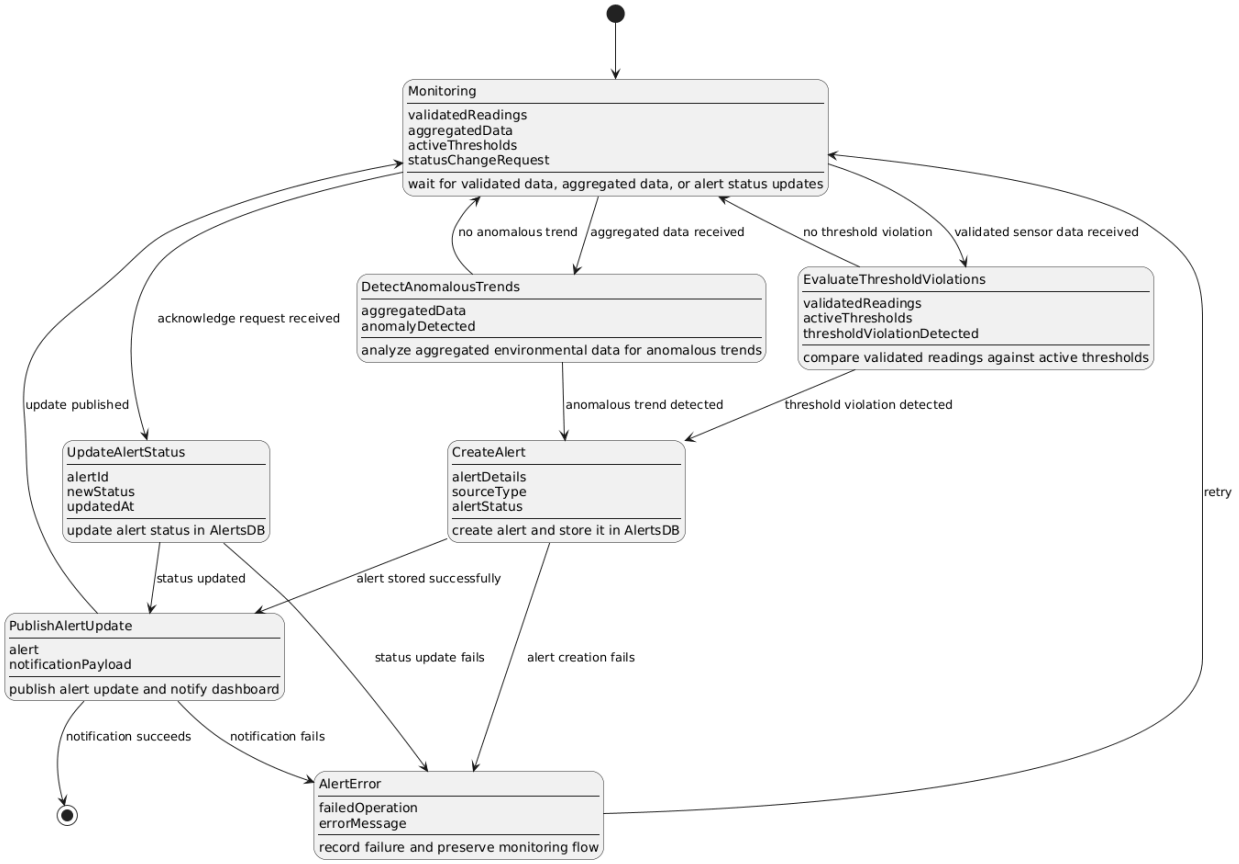

Figure 3: Alerts Management State Chart

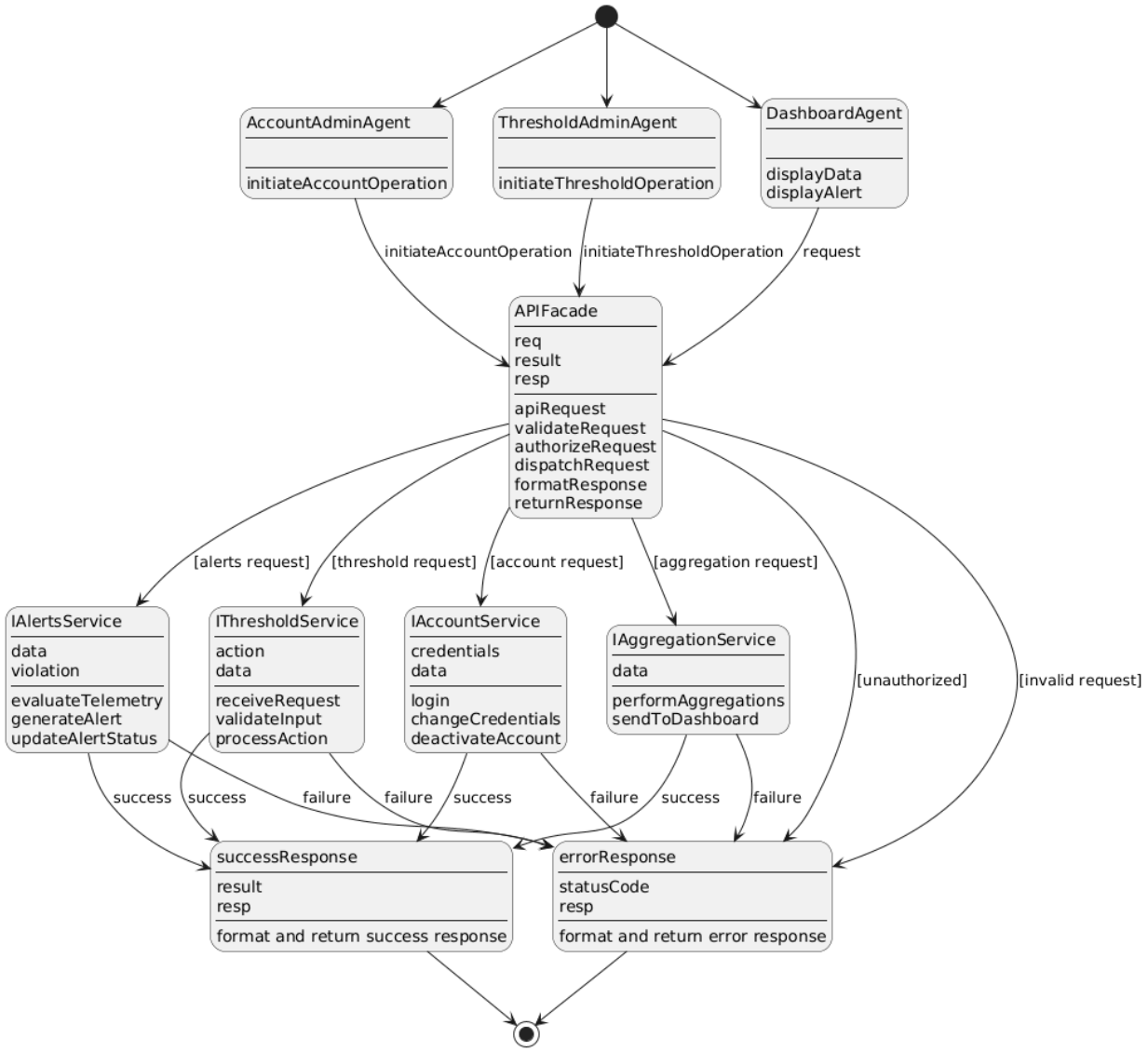

Figure 4: API Facade State Chart

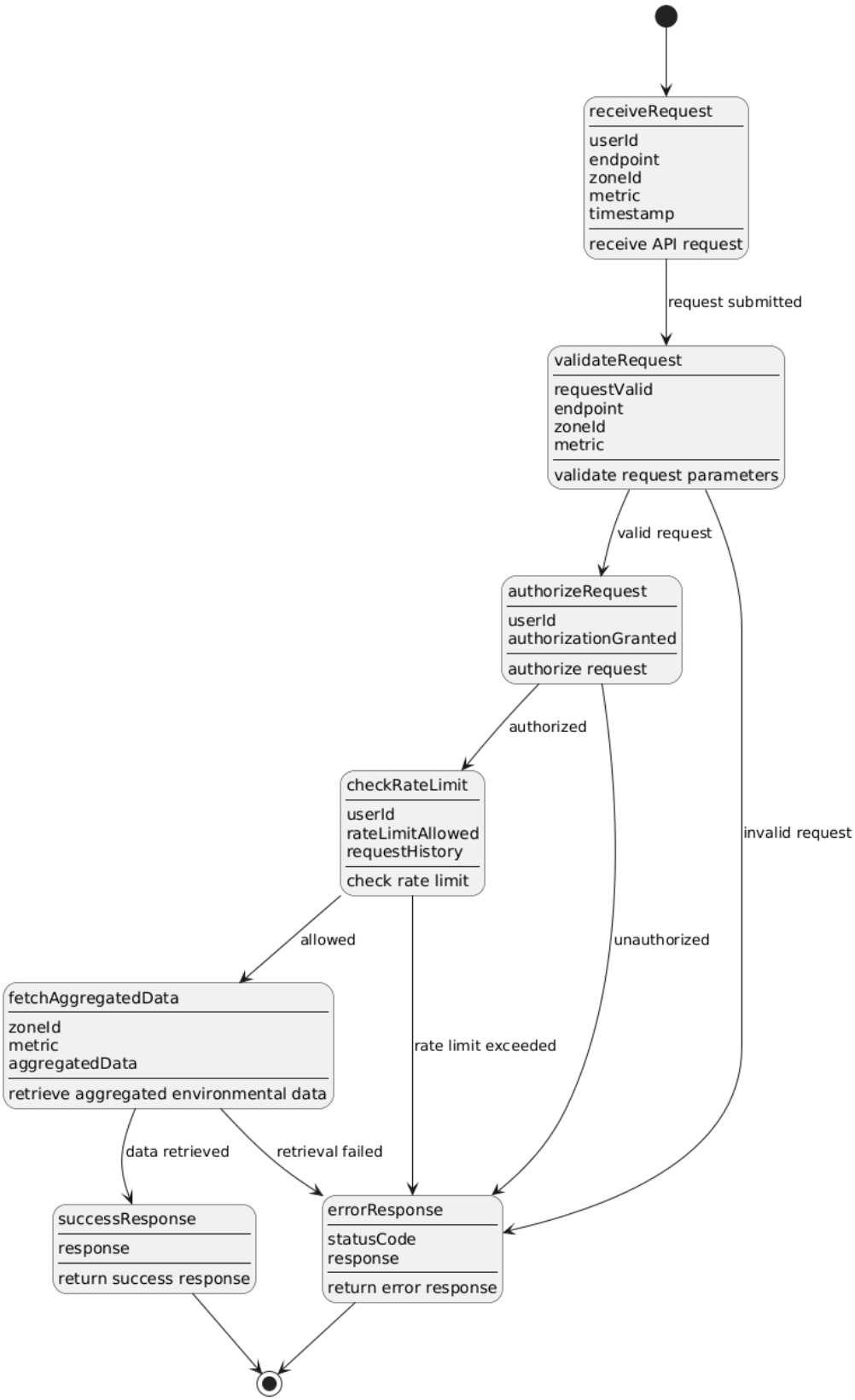

Figure 5: API Management State Chart

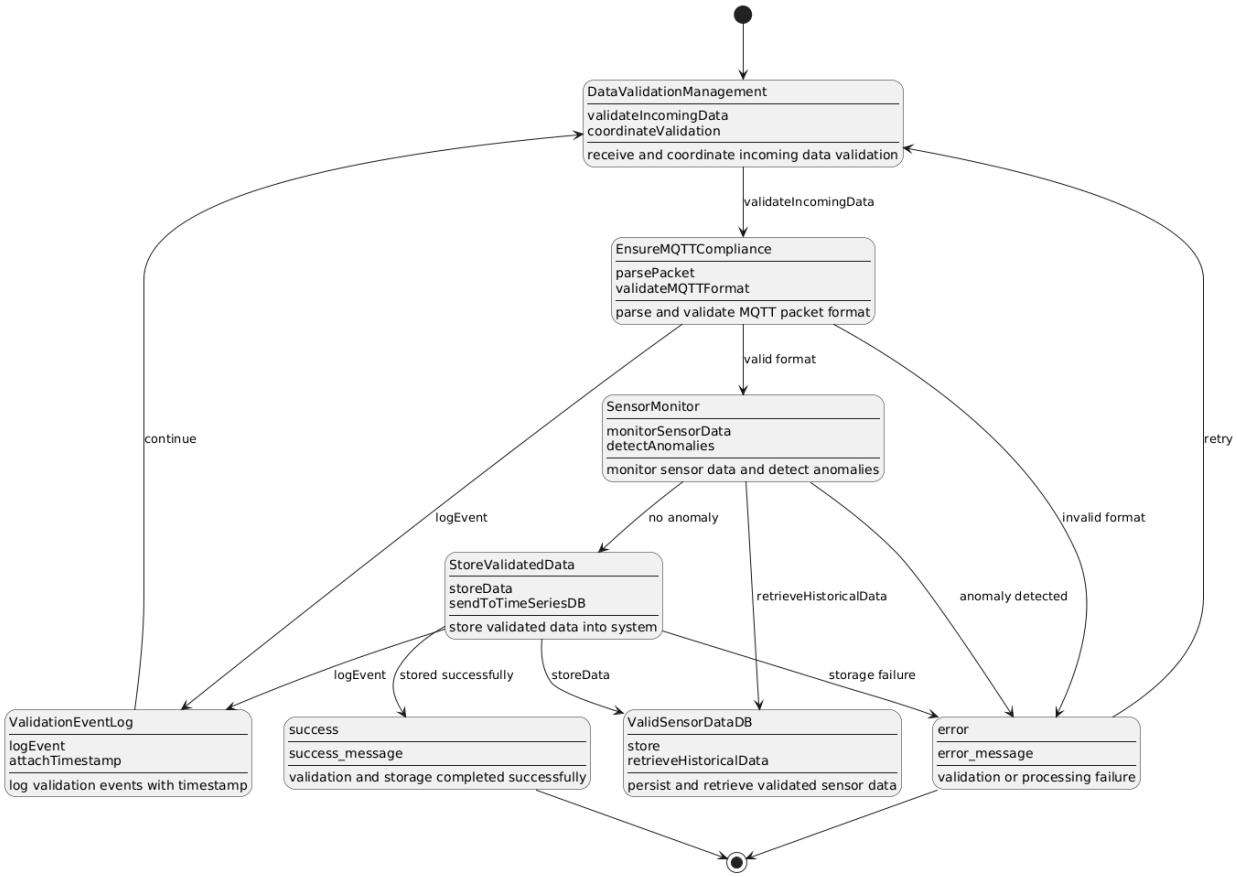

Figure 6: Data Validation Management State Chart

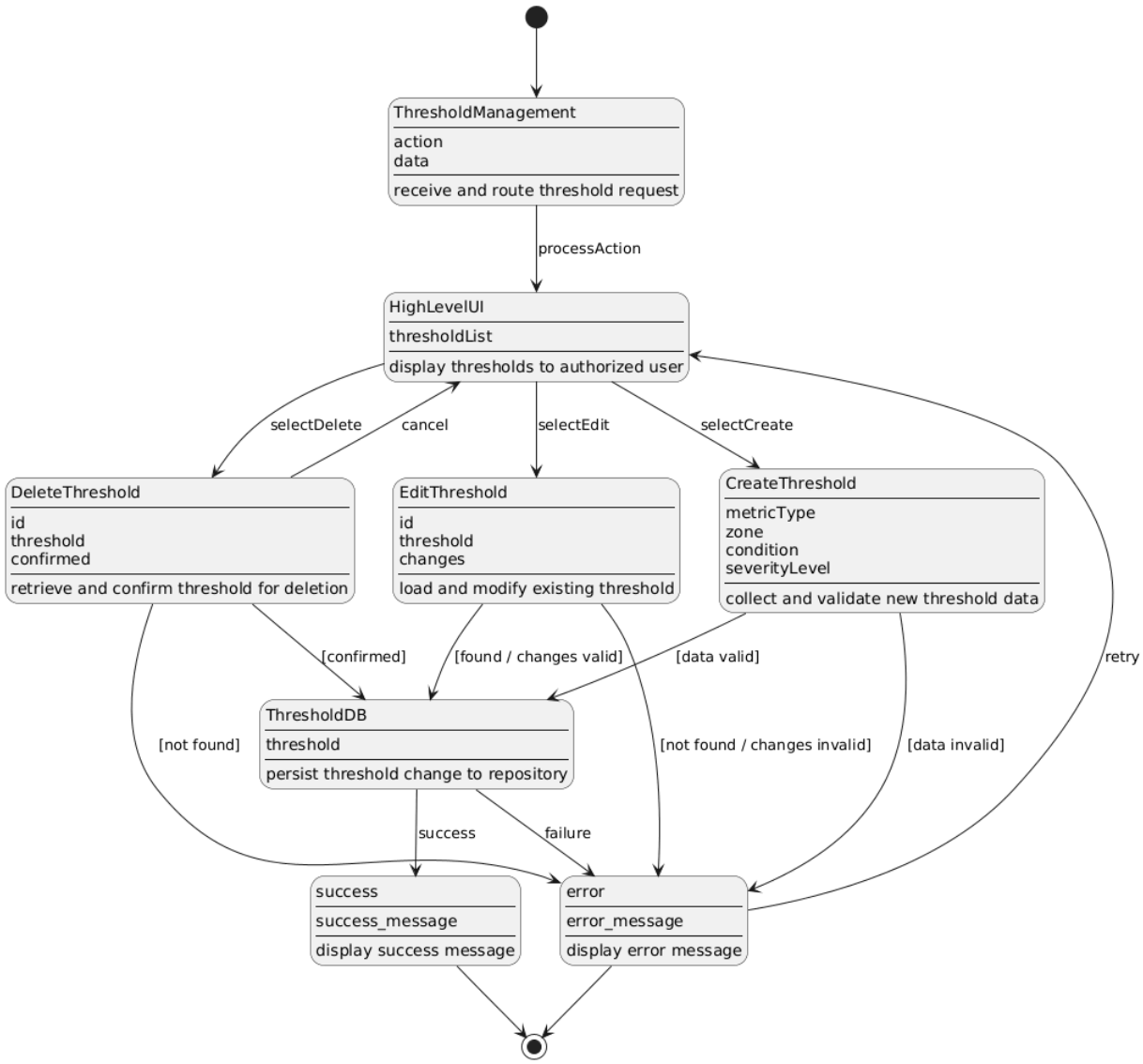

Figure 7: Threshold Management State Chart

3 Sequence Diagrams

This section should provide a sequence diagram for each use case of your application.

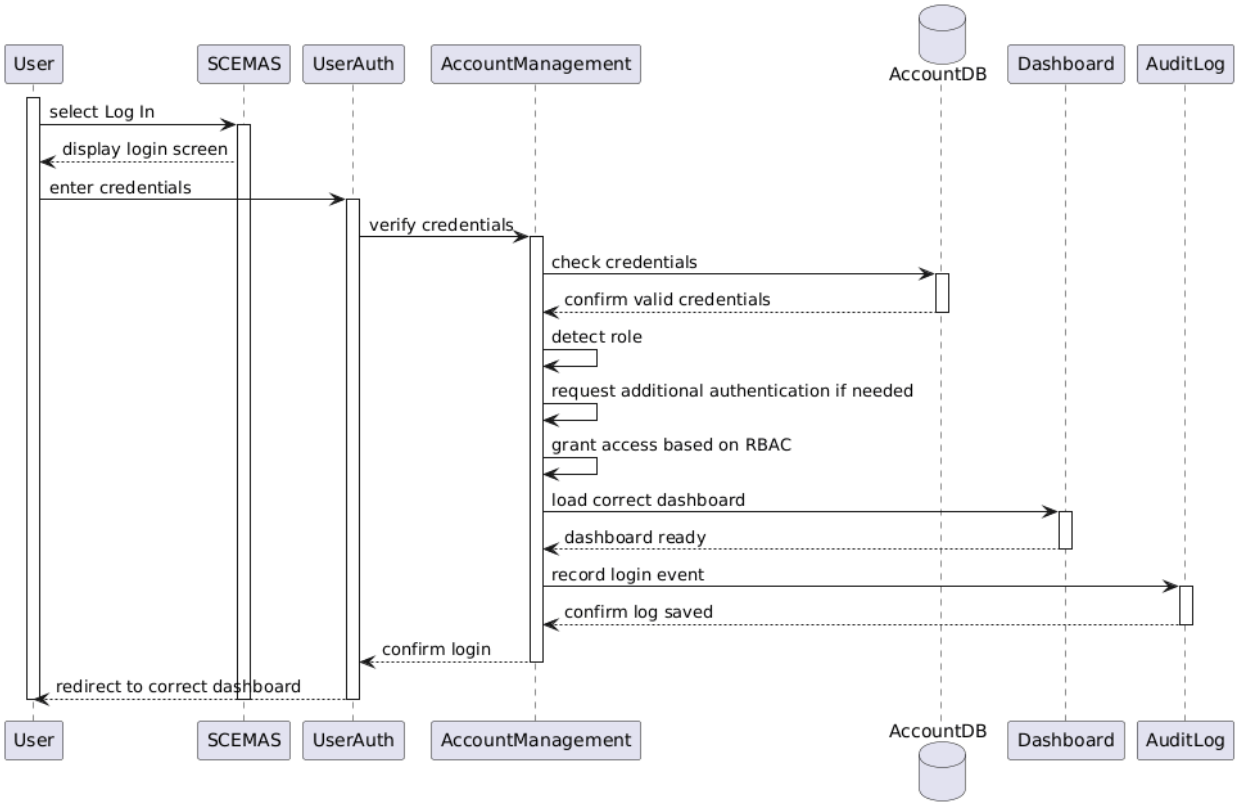

Figure 8: Sequence Diagram for BE1 - Login

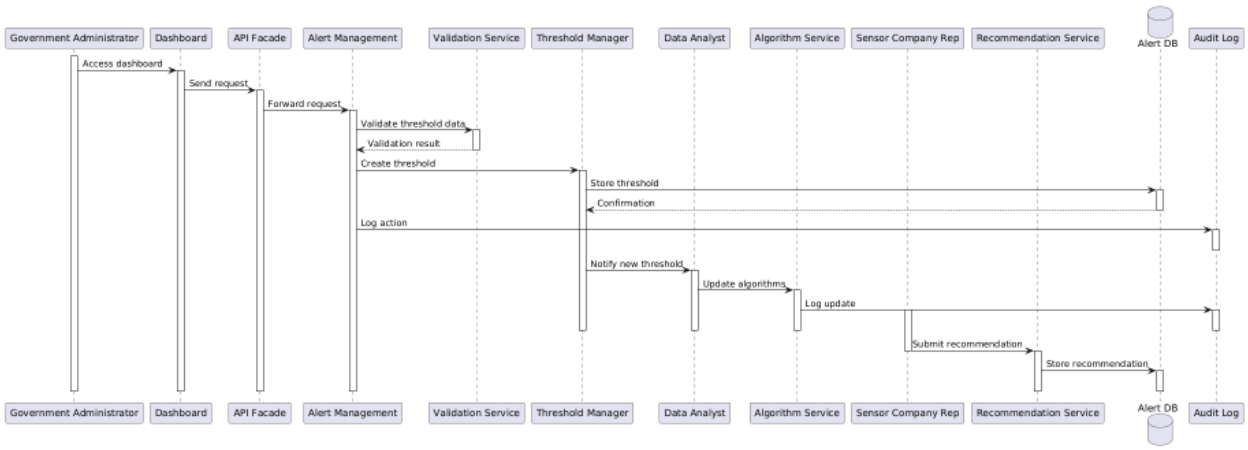

Figure 9: Sequence Diagram for BE2 - Define Alert Thresholds

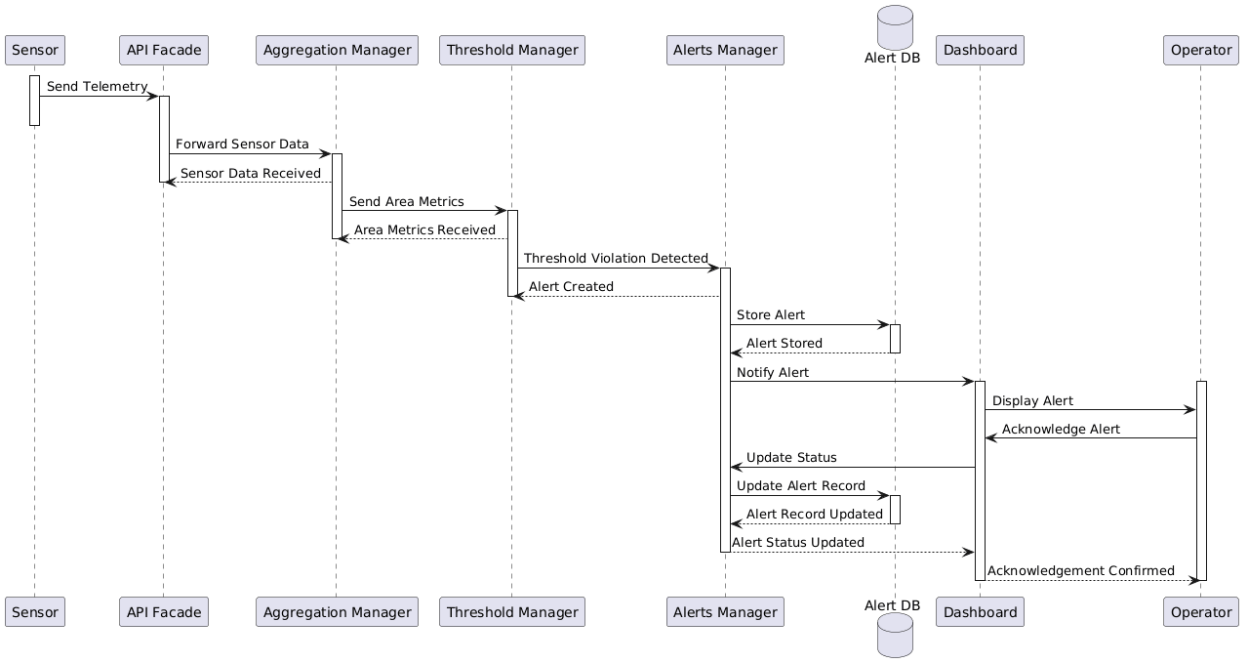

Figure 10: Sequence Diagram for BE3 - Threshold Violations

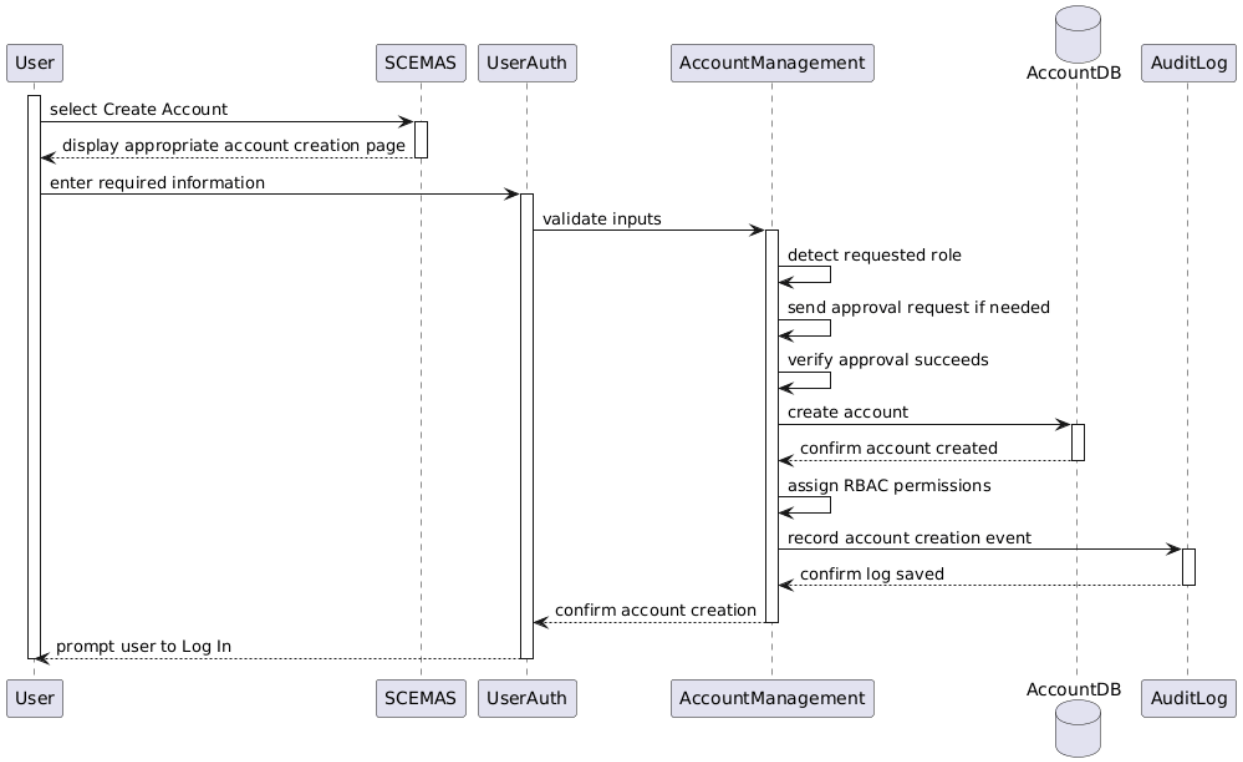

Figure 11: Sequence Diagram for BE4 - Create Account

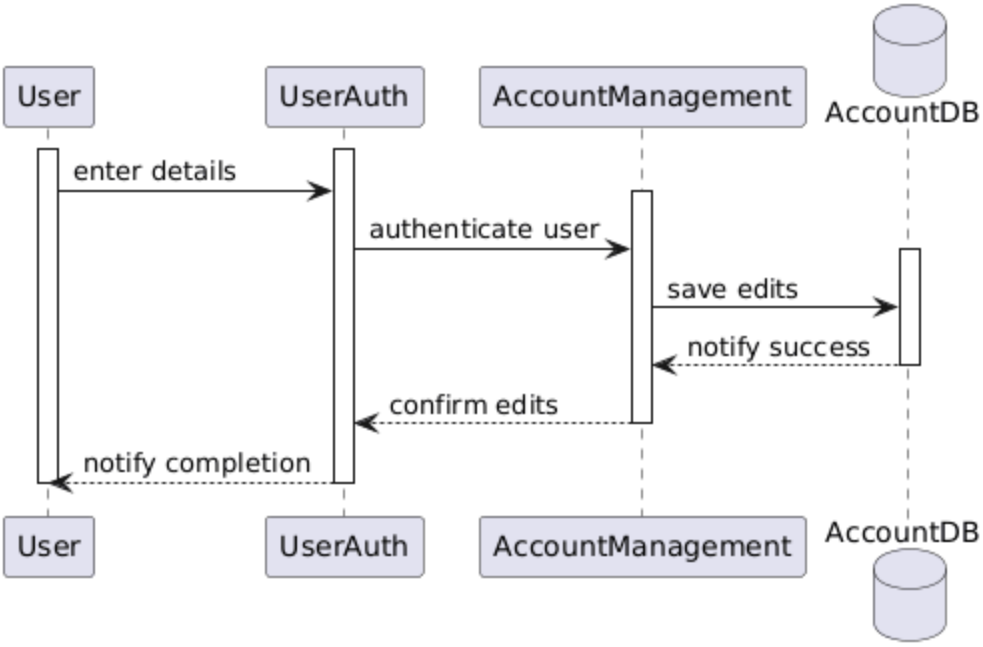

Figure 12: Sequence Diagram for BE5 - Edit Account

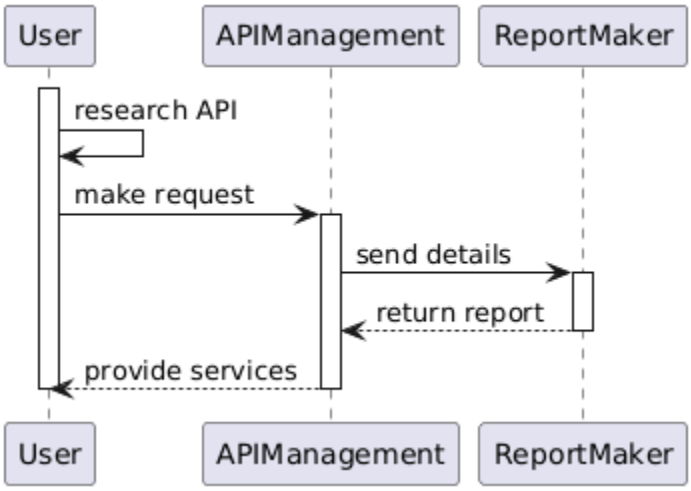

Figure 13: Sequence Diagram for BE6 - Handling API Requests

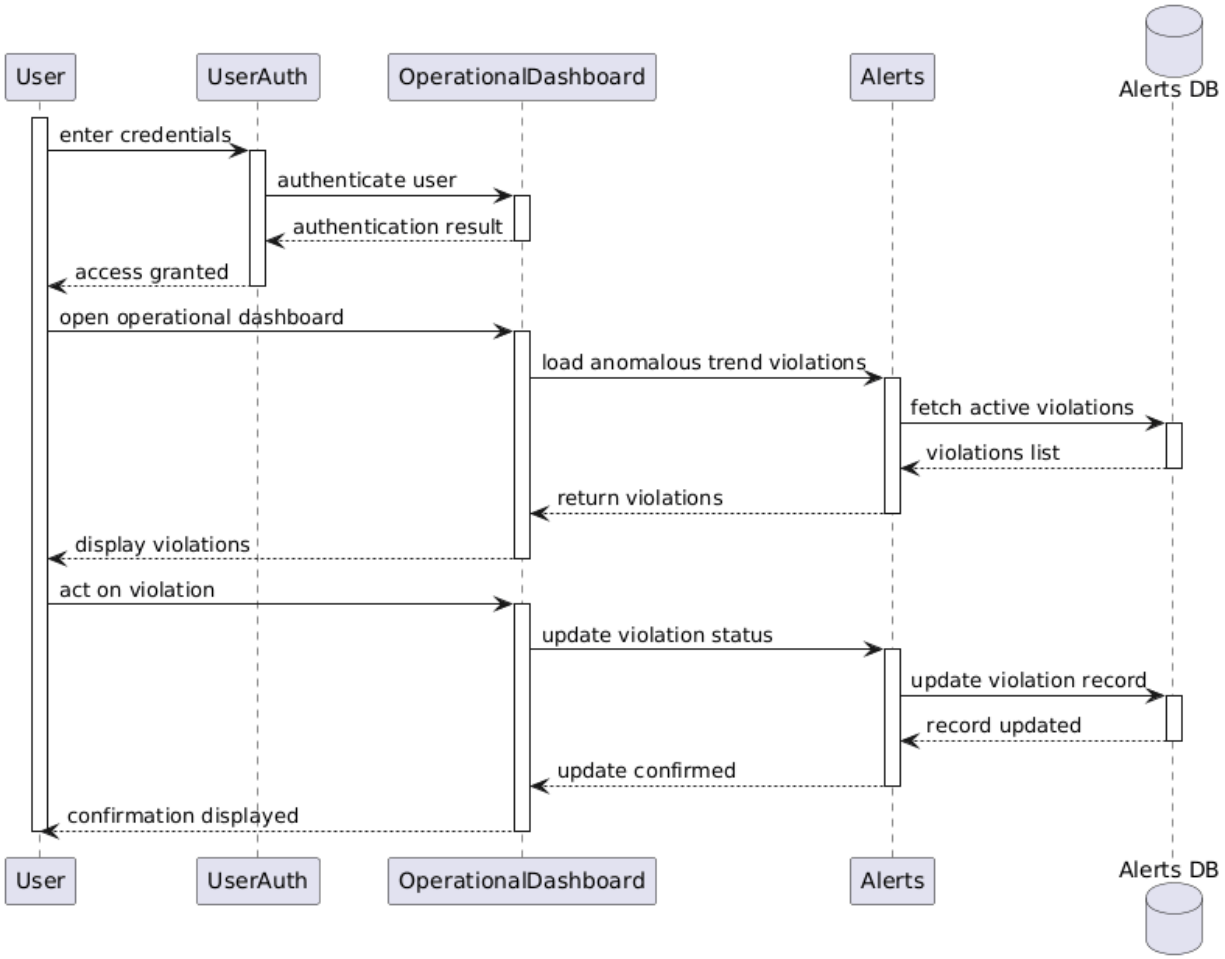

Figure 14: Sequence Diagram for BE7 - Anomalous Trend Violations

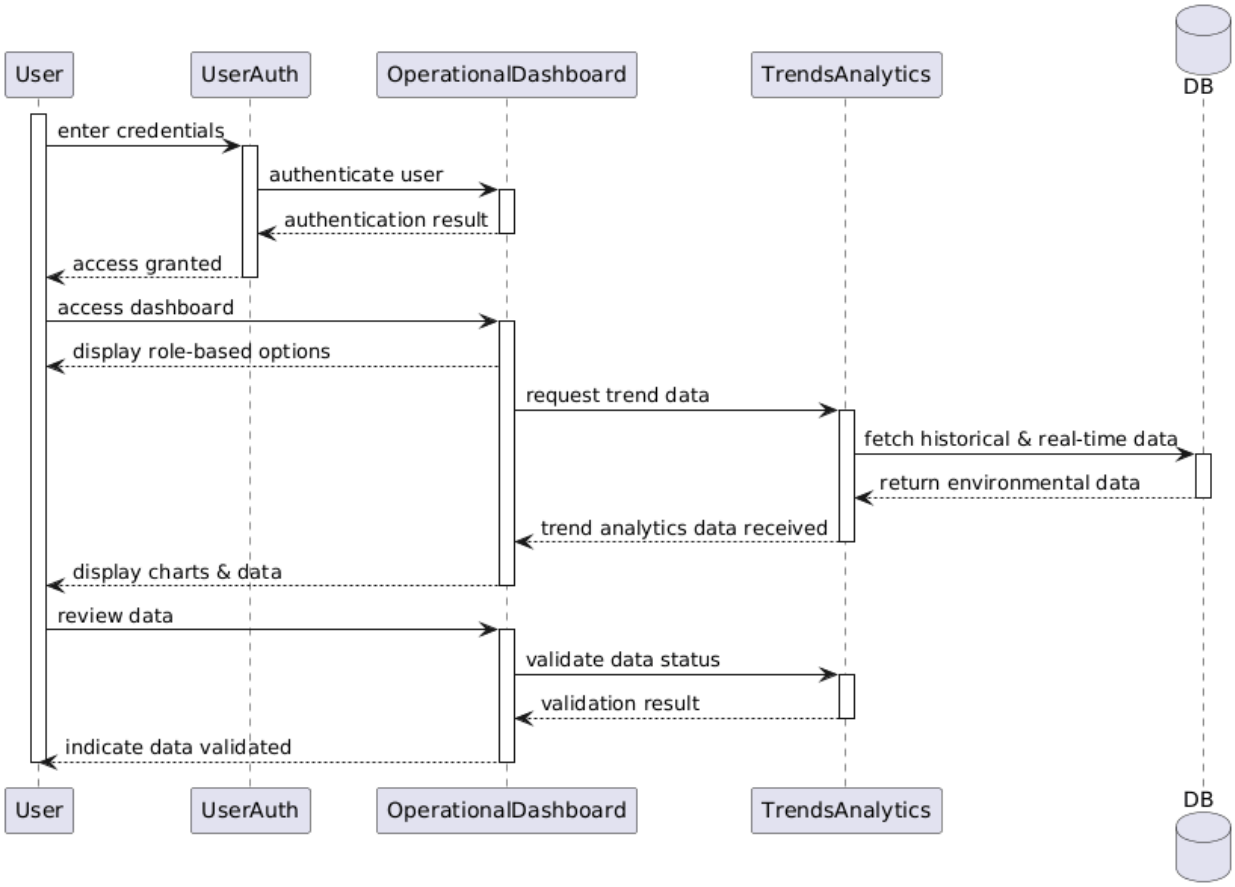

Figure 15: Sequence Diagram for BE8 - Validating Data

4 Detailed Class Diagram

This section should provide a detailed class diagram for your application.

The combined class diagram is shown below:

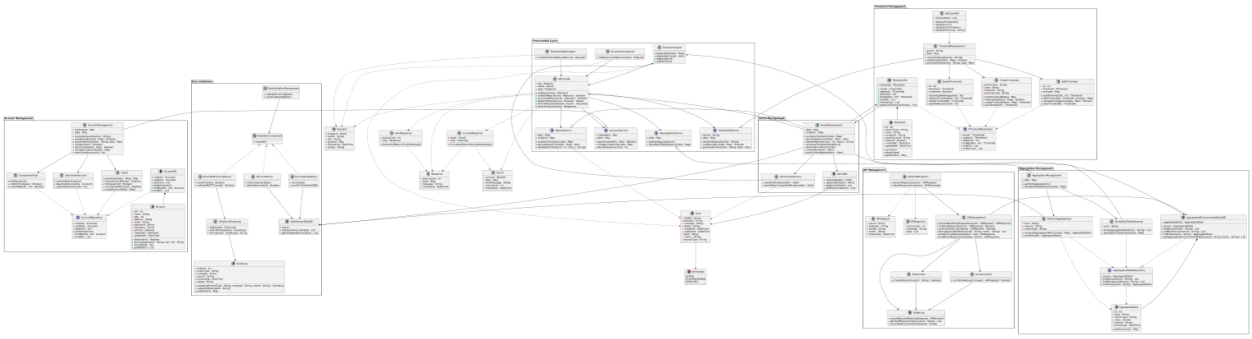

Figure 16: SCEMAS Class Diagram

The class diagram for each subsystem is shown below for better viewing of the connections between different classes.

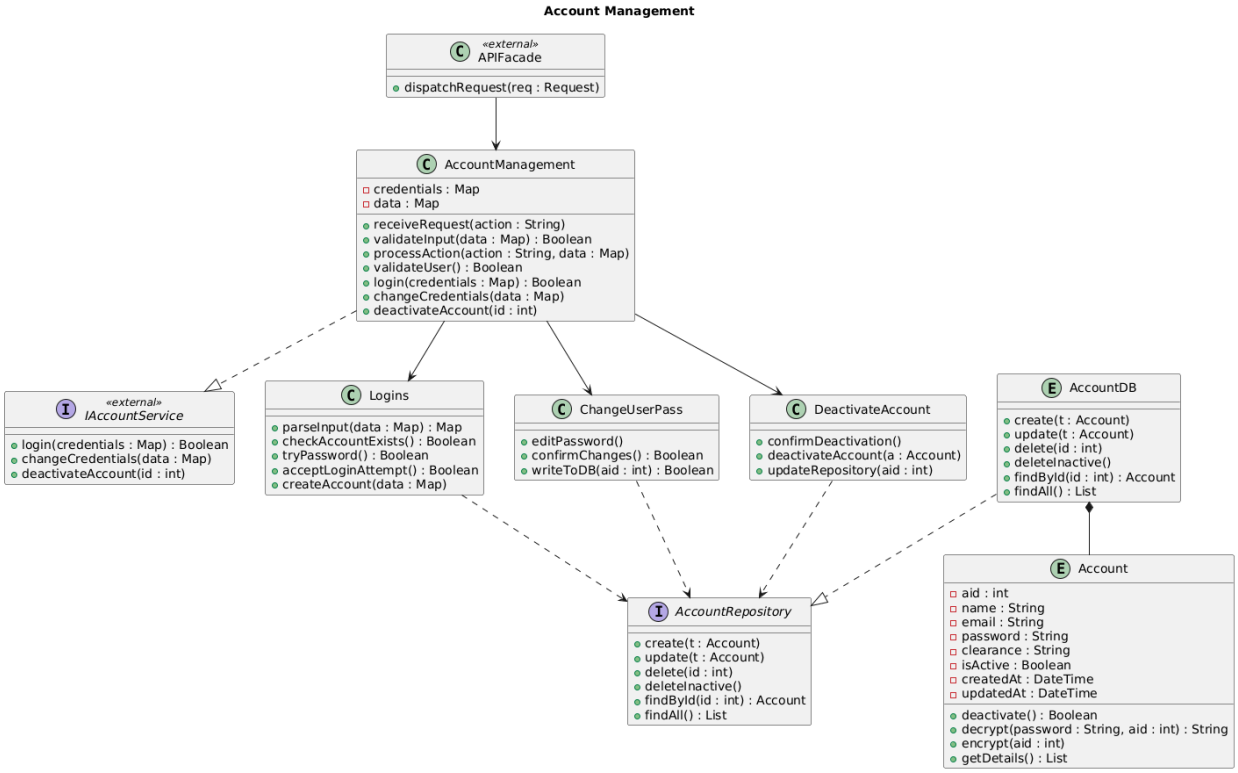

Figure 17: Account Management Subsystem Class Diagram

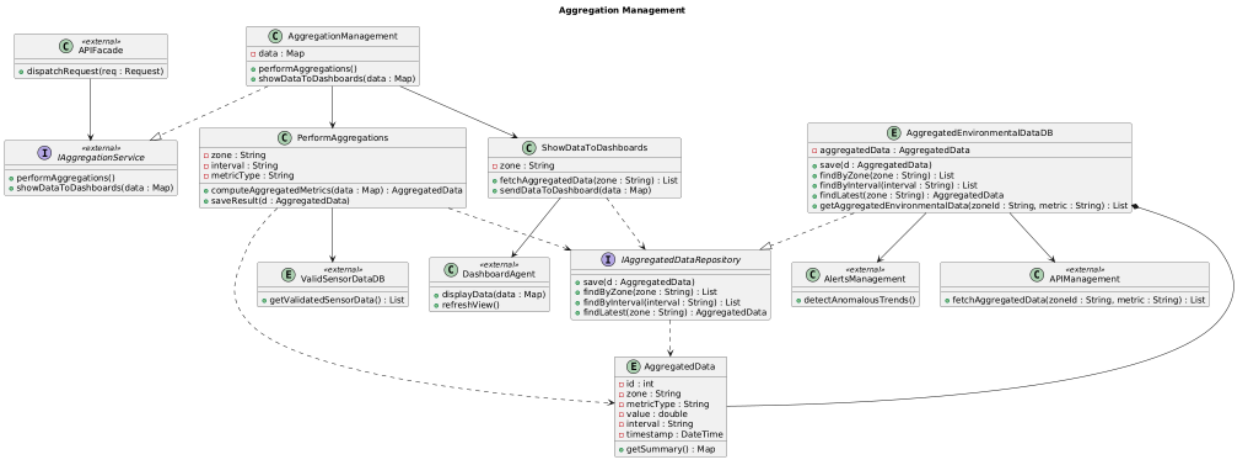

Figure 18: Aggregation Management Subsystem Class Diagram

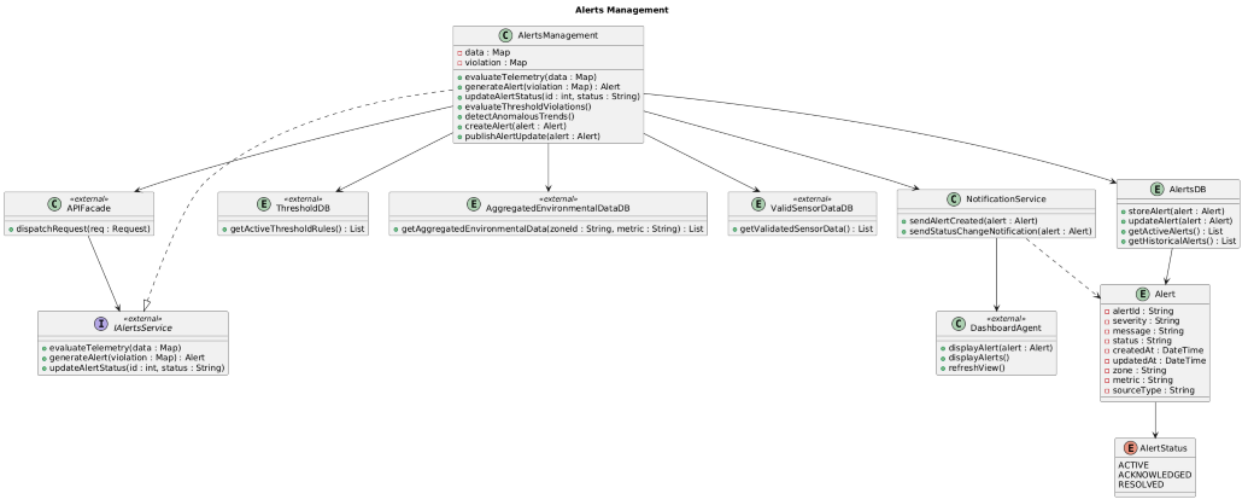

Figure 19: Alerts Management Subsystem Class Diagram

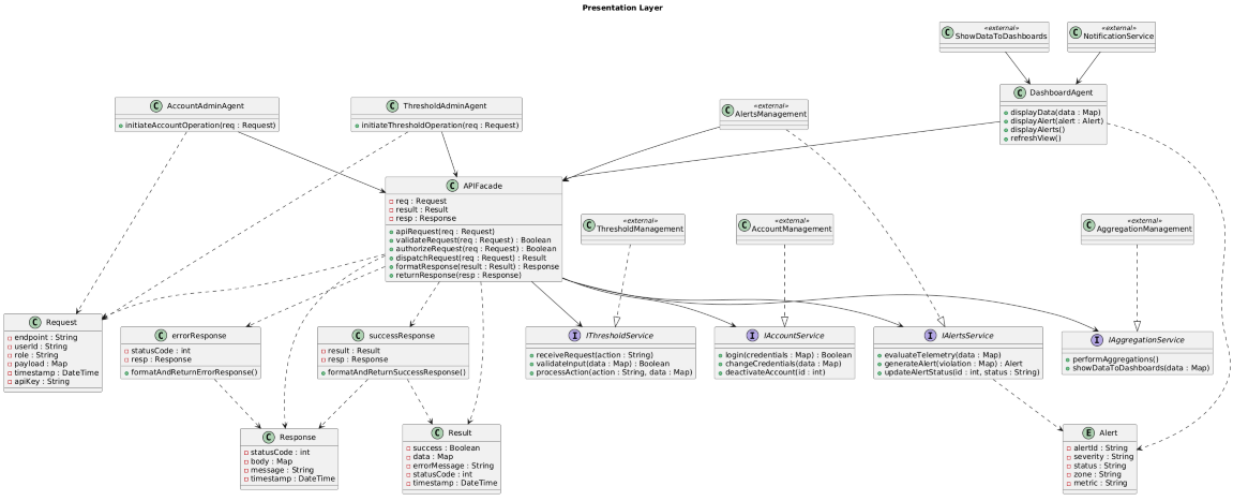

Figure 20: API Facade (Presentation Layer) Subsystem Class Diagram

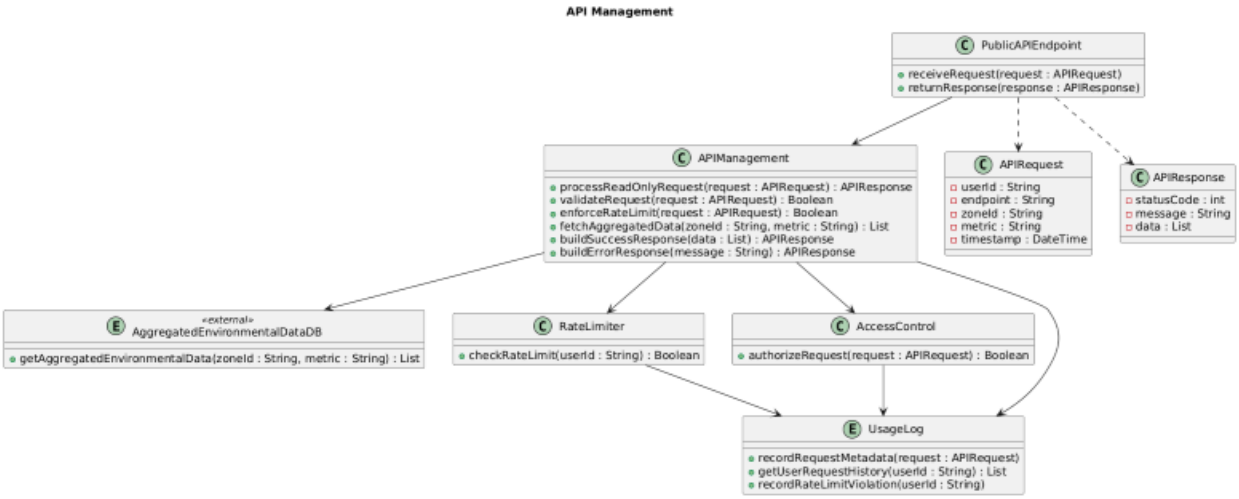

Figure 21: API Management Subsystem Class Diagram

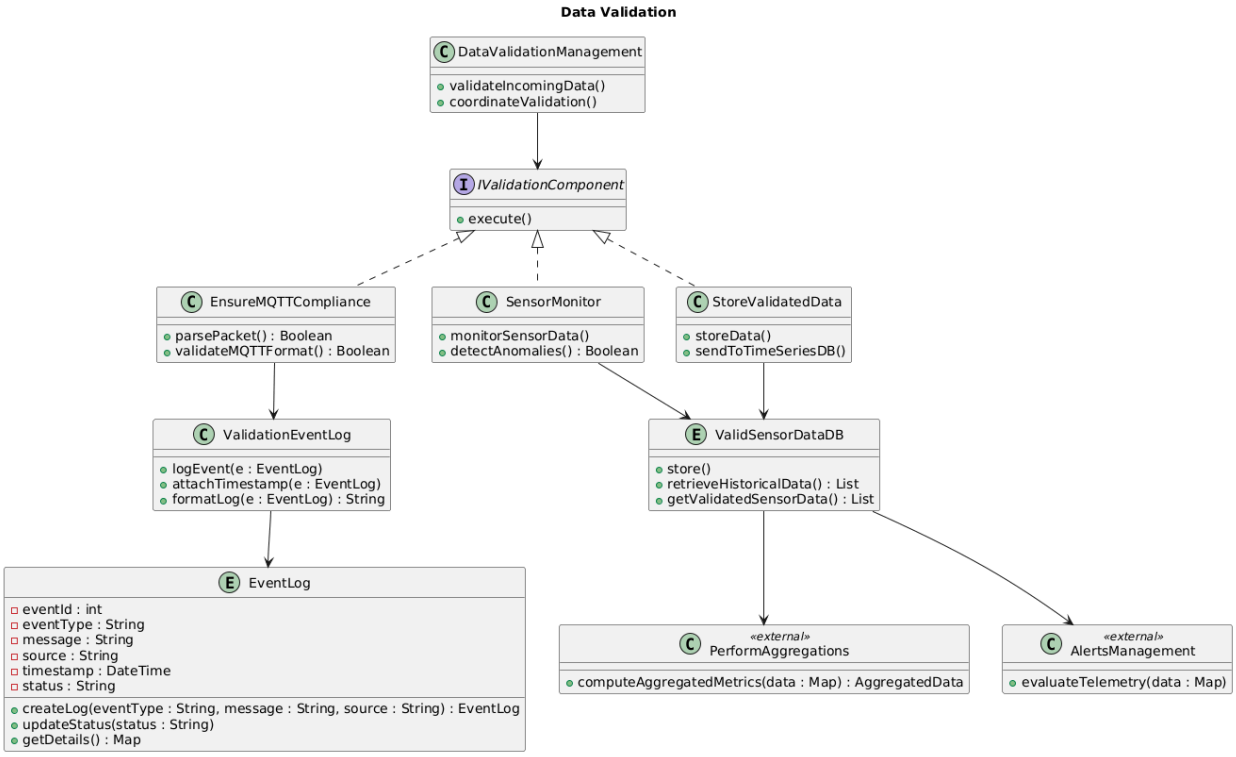

Figure 22: Data Validation Subsystem Class Diagram

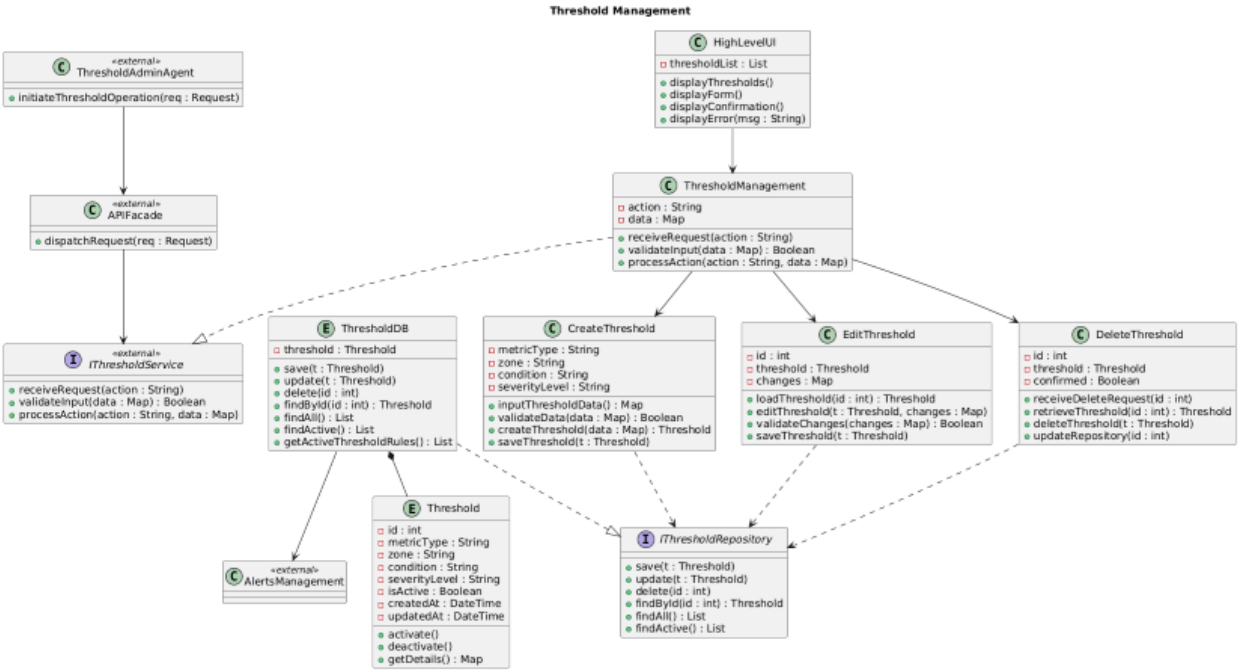

Figure 23: Threshold Management Subsystem Class Diagram

A Division of Labour

Include a Division of Labour sheet which indicates the contributions of each team member. This sheet must be signed by all team members

Aakash Satishkumar:

- Created state charts for API Management and Alerts Management.
- Created subsystem class diagram for API Management and Alerts Management.
- Created sequence diagram for Business Events 1 and 4.

Ali Shareeff:

- Created state charts for Data Validation Management.
- Created subsystem class diagram for Data Validation Management.
- Created sequence diagram for Business Events 7 and 8.

Harmanpreet Singh Sagar

- Created GitHub repository.
- Created state charts for Threshold Management, Aggregation Management, and API Facade.
- Created subsystem class diagram for Threshold Management and API Facade.
- Created sequence diagram for Business Event 3.
- Reviewed all state charts, sequence diagrams, and subsystem class diagrams.
- Created overall SCEMAS class diagram.
- Compiled all PlantUML code to generate images to be inserted into this document.
- Inserted all images into the document.
- Formatting the document.

Jason Kim:

- Created state charts for Account Management.
- Created subsystem class diagram for Account Management.
- Created sequence diagram for Business Events 5 and 6.
- Wrote Section 1.1, 1.2, 1.3

Praneet Singh

- Created subsystem class diagram for Aggregation Management.
- Created sequence diagram for Business Event 2.